

MSc in FM/MTF/QF

Foundations of Econometrics GROUP COURSEWORK

Academic Year 2025/2026, Fall 2025

Question 1: Descriptive Analysis and Stylised Facts

You are a junior quant analyst at a global asset management firm. Before conducting any econometric modelling, your manager asks you to carry out a diagnostic descriptive analysis of the firm's five core risk factors. This step is crucial for understanding the **behaviour of the data**, identifying potential pitfalls, and ensuring model assumptions are appropriate.

The assets to analyse (daily frequency: 1 January 2000–30 August 2025):

- An equity index (FTSE 100 or S&P 500)
- A government bond yield (U.S. or U.K.)
- A corporate bond spread (U.S. or U.K.)
- A commodity (e.g., oil, gold, copper)
- A major exchange rate (e.g., GBP/USD, EUR/USD, USD/JPY)

1. Exploratory Analysis

- Download and plot the time series of prices/levels and returns (log-differences where appropriate).
- Provide summary statistics (mean, variance, skewness, kurtosis, correlations) for each asset's returns.

2. Stylised Facts Identification

- Examine your plots and statistics to highlight whether the data exhibit well-known financial time series features:
 - Volatility clustering
 - Fat tails (leptokurtosis)
 - Leverage/asymmetry effects
 - Mean reversion (in yields or spreads)
 - Persistence in volatility
- Provide at least one visual illustration (e.g., returns vs. squared returns, volatility spikes during crises).

3. Practitioner's Interpretation

- Relate the stylised facts you identify to real-world events (e.g., Global Financial Crisis, sovereign debt crisis, COVID-19 pandemic, commodity shocks).
- Explain why recognising these stylised facts is important for risk modelling, portfolio allocation, and forecasting.

4. Professional Communication. Draft a short non-technical note (max 100 words) to your manager explaining the two most important stylised facts in your dataset, and why they matter for the firm's trading and risk strategies.

QUESTION 2. Unit Roots in Asset Prices

You have just joined a **quant research team** at a global asset management firm. Your manager asks you to check whether the firm's key risk factors behave as random walks or whether they are mean-reverting. This is crucial for portfolio modelling, forecasting, and risk control.

The assets to analyse (daily data aggregated to quarterly, 2000–2025):

- An equity index (FTSE 100 or S&P 500)
- A government bond yield (U.S. or U.K.)
- A corporate bond spread (U.S. or U.K.)
- A commodity (e.g., oil, gold, copper)
- A major exchange rate (e.g., GBP/USD, EUR/USD, USD/JPY)

1. Empirical Testing

- Aggregate each series to quarterly frequency.
- Perform unit root tests. Report the test statistics and conclusions for both levels and first differences.

2. Practitioner's Interpretation

- Explain what your results imply about market efficiency: do prices follow a martingale/random walk, or is there evidence of mean reversion?
- Identify one asset in your sample where the result was surprising. Suggest a market-based reason (e.g., crisis periods, monetary policy interventions, structural breaks).

3. Decision-Making

- As a quant, advise whether each series should be modelled in levels or returns when building forecasting models and risk systems.
- Discuss the risks of misclassifying a non-stationary process as stationary (or vice versa) for portfolio construction and risk management.

4. Professional Communication. Draft a short executive summary (max 100 words) to brief the Chief Risk Officer. Keep it non-technical: explain why stationarity matters and what the practical implications are for investment strategy.

QUESTION 3: Cointegration in a Quant Analyst Role

You are working as a quantitative analyst at a hedge fund that trades multi-asset portfolios. Your manager asks you to investigate whether there are **long-run equilibrium relationships** among the following five assets (daily data aggregated to quarterly, 2000–2025):

- A stock index (FTSE 100 or S&P 500)
- A government bond yield (U.S. or U.K.)
- A corporate bond spread (U.S. or U.K.)
- A commodity (e.g., Brent oil, gold, copper)
- A major exchange rate (e.g., GBP/USD, EUR/USD, USD/JPY)

1. Empirical Testing

- Test for cointegration among the five series, using the stock index as dependent variable.
- Report whether a cointegration relationship exists, and if so, estimate the cointegration vector.
- Test whether the two markets (the US and UK government bond yield series represent) are integrated rather than converging or diverging.

2. Practitioner's Interpretation

- Explain what the cointegration vector means in portfolio terms (e.g., a “hedge ratio” or “long-run pricing spread”).
- Identify one episode in your sample where the equilibrium broke down temporarily (plot the deviation). Link this to an actual event (e.g., Global Financial Crisis, COVID-19, commodity price shock, monetary policy shift).

3. Decision-Making

- Imagine your manager is considering a pairs-trade / relative-value trade based on your cointegration results.
- Would you recommend this strategy? Why or why not?
- Discuss risks: structural breaks, transaction costs, model misspecification, or regulatory changes.

4. Professional Communication. Summarise your findings in a short “trader’s note” (max 100 words) that your manager could use to brief the trading desk.

QUESTION 4: Spurious Regression in Multi-Asset Analysis

You are a quant analyst at a macro hedge fund. One of your portfolio managers claims to have discovered a “strong predictive relationship” between equity prices and bond yields that could be used for trading signals. You suspect this may be a case of **spurious regression**, because both series are trending.

Your data (daily data, aggregated to quarterly, 2000–2025):

- An equity index (FTSE 100 or S&P 500)
- A government bond yield (U.S. or U.K.)
- A corporate bond spread (U.S. or U.K.)
- A commodity (oil, gold, or copper)
- An exchange rate (e.g., GBP/USD, EUR/USD, USD/JPY)

1. Naïve Regression

- Choose one pair of series from the five that appears to trend together.
- Estimate a regression of one level on the other (include a constant and, if needed, a deterministic trend). Report R^2 , t-statistics, and the Durbin–Watson statistic.

2. Diagnostic Check

- Test the regression residuals for a unit root.
- Explain why a “significant” regression with a high R^2 may still be misleading.

3. Re-specification

- Re-estimate the relationship in first differences/returns.
- Compare the results to the levels regression and comment on the difference.

4. Practitioner’s Reflection

- As a quant, explain to your portfolio manager:
 - Why the original regression was spurious.
 - The risks of basing trading strategies on such relationships (false predictability, mis-estimated hedge ratios, mispriced risk).
 - How pre-testing for unit roots and cointegration should be built into the research process.

5. Professional Communication. Write a short trader’s warning note (max 100 words) summarising why spurious regressions can generate false signals in cross-asset trading.

QUESTION 5: Granger Causality and Price Discovery

You are a quantitative analyst at a multi-asset trading desk. The head of trading believes that understanding which markets lead others in processing information could improve **forecasting** and hedging strategies. She asks you to investigate the direction of information flows across asset classes.

Your dataset (daily data, aggregated to weekly, 2000–2025):

- An equity index (FTSE 100 or S&P 500)
- A government bond yield (U.S. or U.K.)
- A corporate bond spread (U.S. or U.K.)
- A commodity (e.g., oil, gold, or copper)
- An exchange rate (e.g., GBP/USD, EUR/USD, USD/JPY)

1. Empirical Analysis

- Aggregate daily data to weekly end-of-week values.
- Perform pairwise **Granger causality tests**.

2. Practitioner's Interpretation

- Identify at least one case where asset A helps forecast asset B.
- Explain why this result makes sense economically and highlight one result that was unexpected. Link both to events in your sample (e.g., GFC, sovereign debt crisis, commodity shocks, monetary policy shifts).

3. Practitioner's Implications

- As a quant, recommend how the desk could use these results in:
 - Forecasting returns or volatility.
 - Cross-asset hedging strategies.
 - Understanding information transmission in crises.

4. Risk Assessment

- Highlight limitations of Granger causality for trading (e.g., sensitivity to lag choice, structural breaks, short-lived lead–lag effects).
- Suggest one robustness check (e.g., subsample analysis, rolling-window estimation).

5. Professional Communication. Draft a one-paragraph briefing (max 100 words) for the head of trading, summarising your main finding and its implications for price discovery across the five assets.

QUESTION 6: Volatility Modelling with GARCH

(TBC)

You are a quantitative analyst on the risk management team of a global investment bank. Your manager asks you to evaluate the **volatility dynamics** of different asset classes to improve Value-at-Risk (VaR) and stress-testing frameworks.

Your dataset (daily returns, 1 January 2000–30 August 2025):

- An equity index (FTSE 100 or S&P 500)
- A government bond yield (U.S. or U.K.)
- A corporate bond spread (U.S. or U.K.)
- A commodity (e.g., oil, gold, or copper)
- An exchange rate (e.g., GBP/USD, EUR/USD, USD/JPY)

1. Model Estimation

- For each asset, estimate at least three GARCH-type models:
 - GARCH(1,1)
 - EGARCH(1,1)
 - GJR-GARCH(1,1)
- Fit each model under Gaussian and Student-t error distributions.

2. Model Selection & Diagnostics

- Use information criteria and residual diagnostics (ACF of standardized residuals, ARCH-LM tests) to select the best-fitting model for each asset.
- Provide at least one residual plot that demonstrates why your chosen model fits better than the alternatives.

3. Practitioner's Interpretation of Volatility Dynamics

- Identify whether volatility is highly persistent, whether leverage/asymmetry effects are present, and whether fat tails matter.
- Highlight one crisis period (e.g., 2008, 2011 sovereign debt crisis, 2020 COVID shock) and explain how volatility dynamics in your models capture it.

4. Risk Management Implications

- Using the best-fitting model for one of your assets, generate a 1-day ahead volatility forecast during a high-volatility episode.
- Explain how such forecasts would be used for:
 - Setting daily VaR limits.
 - Pricing options or derivatives.
 - Stress testing during turbulent periods.

5. Professional Communication: Draft a short internal memo (max 100 words) to the Chief Risk Officer, summarising (1) the main volatility features you found, (2) why symmetric GARCH may be insufficient, and (3) how the results impact portfolio risk monitoring.

Notes to Students

- You must download, clean, and analyse the data yourself; all plots, tables, and results must be included in the report.
- Answers should combine empirical results with interpretation, linking statistical and econometric evidence to financial reasoning.
- AI-generated outputs that are not grounded in your dataset will not be sufficient, as marks will be awarded for data preparation, numerical outputs, contextual analysis, and interpretation.
- Emphasis will be placed on your ability to reason as a professional quant analyst rather than on formulaic results.
- It is compulsory to use OxMetrics9 to execute the empirical applications. Moreover, it is advisable that all empirical applications are replicated in Python.
- The deadline for the **group coursework** is **4:00 p.m. on Friday, 28 November 2025**. Submissions must be uploaded to **Moodle** and must include the following:
 - **A PDF file** of the coursework report (1,500–2,000 words, max 8 pages, including references/appendices).
 - **A zipped folder** containing:
 - An Excel file with the downloaded dataset and any transformations applied.
 - The Python code (if used) required to replicate your empirical results.
 - Please ensure that your Excel file is sufficient to replicate the reported results in OxMetrics and/or replicated in Python.

Giovanni Urga, 30 September 2025